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SECTION: G

## Lab 3: Architectural Pattern Selection — Written Justification

We chose Microservices Architecture for the Freelance Content Creator Escrow Platform.

### Architectural Choice

The platform is decomposed into independently deployable services — Contract & Milestone Manager, Watermarking Service, Payment & Escrow Service, and Notification Service — each backed by a shared Database & Audit Log component, communicating through well-defined interfaces exposed to the User Interface and to the external Payment Gateway.

### Two Reasons

1. Independent scaling of distinct workloads: Watermarking (FR-004) is a compute-heavy, latency-sensitive task (NFR-001 requires under 5 seconds), while payment processing (UC-08/UC-09) and notifications (FR-005) have very different load and reliability profiles. Microservices let each service scale independently instead of forcing one monolithic layer to handle all three.
2. Fault isolation for a money-handling system: If the external Payment Gateway is slow or the Watermarking Service fails, the escrow platform must still let creators submit drafts and sponsors leave review comments. Isolating each concern into its own service means a failure in one (e.g., alternate flow 4a - payment declined) does not bring down draft submission, review, or notifications.

### Security Advantage

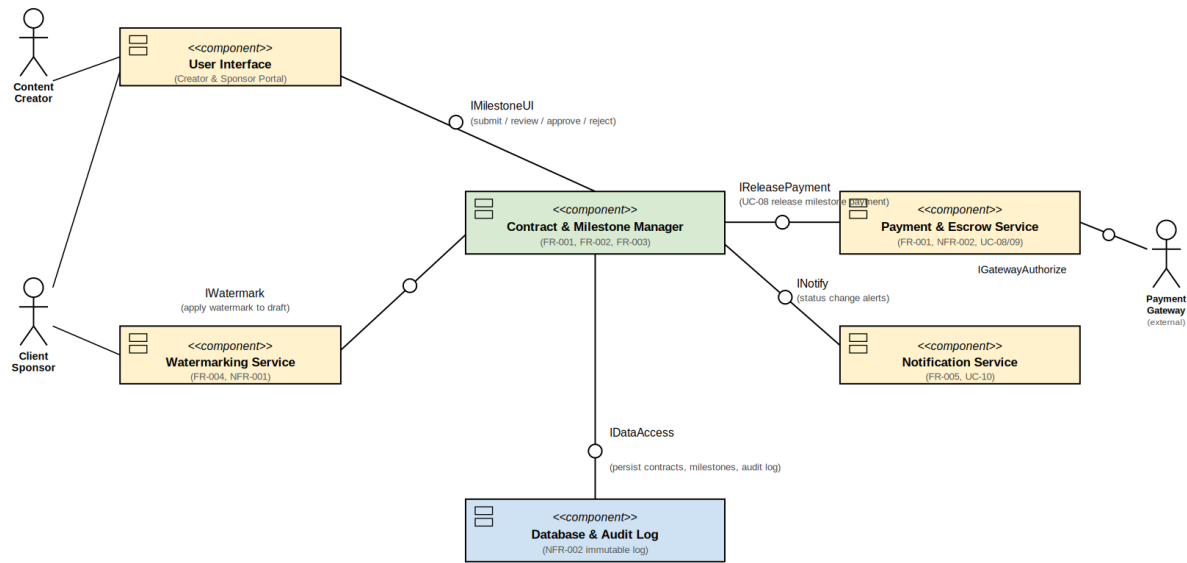
Isolating the Payment & Escrow Service as its own microservice confines all TLS-encrypted transactions and audit-log writes (NFR-002) to a single, narrowly-scoped component with its own access controls, rather than exposing payment logic inside a shared layer that also handles UI or notification code — reducing the attack surface around escrow funds and simplifying compliance auditing.

### Performance Benefit

Because the Watermarking Service is deployed and scaled independently of the rest of the system, it can be given dedicated compute resources to consistently meet the sub-5-second watermarking requirement (NFR-001) even during peak load, without competing for resources with milestone management, payment, or notification traffic.

Freelance Content Creator Escrow Platform — UML Component Diagram

(Microservices Architecture)



○ Provided Interface (ball)